

University Academic Handbook

Bachelor of Science in Computer Science

2026-2027 Academic Year

*Contains all academic policies, course descriptions, graduation rules,
prerequisite relationships, and GPA requirements.*

Program Overview

The Bachelor of Science in Computer Science (BSCS) degree program prepares students for careers in software engineering, data science, artificial intelligence, and computational research. This handbook contains all academic policies, course descriptions, graduation rules, and GPA requirements governing the BSCS program. Students must satisfy all graduation rules and maintain minimum GPA requirements to earn the degree.

Academic Policies and Regulations

General Academic Standards

All students enrolled in the BSCS program must adhere to the university's academic integrity policies. The minimum GPA requirements apply to all attempted coursework, including repeated courses. Graduation rules mandate that students complete a minimum of 120 credit hours with a cumulative GPA of at least 2.0. Students must also maintain a major GPA of at least 2.5 in all Computer Science courses to satisfy the graduation rules.

Attendance and Participation

Regular attendance is expected in all courses. Instructors may establish specific attendance policies in their syllabi. Students who miss more than 20% of class sessions may be subject to administrative withdrawal. The graduation rules do not permit exceptions for attendance violations.

Degree Requirements and Graduation Rules

Overview of Graduation Rules

The BSCS program enforces strict graduation rules to ensure academic rigor. All graduation rules must be satisfied before a student is eligible to receive the degree. The graduation rules are divided into the following categories:

Requirement Category	Credit Hours	Minimum GPA
General Education	42	2.0
Computer Science Core	48	2.5
Mathematics Foundation	18	2.5
Technical Electives	12	2.0
Free Electives	0	-
Total	120	2.0 Cumulative

Detailed Graduation Rules

The graduation rules for the BSCS degree require students to:

- Complete exactly 120 semester credit hours of approved coursework.
- Satisfy all prerequisite relationships before enrolling in advanced courses.
- Achieve a cumulative GPA of at least 2.0 across all coursework.
- Achieve a major GPA of at least 2.5 in all CS-prefixed courses.
- Complete the senior capstone sequence (CS 4901 and CS 4902).
- Fulfill the university residency requirement of 30 credit hours.
- File a graduation application by the published deadline.

These graduation rules are non-negotiable and apply to all students regardless of entry date. Violations of graduation rules will delay degree conferral.

Graduation Rules for Transfer Students

Transfer students must satisfy the same graduation rules as native students. All transfer credits are evaluated against the graduation rules. Transfer students must complete at least 30 credit hours at this institution and must meet the same GPA requirements as degree-seeking students. The graduation rules for transfer students include additional transcript evaluation procedures.

Core Curriculum and Course Catalog

Computer Science Core Courses (48 Credit Hours)

The Computer Science core establishes the foundational knowledge required for the degree. All core courses carry CS prefixes and are subject to the major GPA requirements.

CS 1101 - Introduction to Programming

Credit Hours: 3

Prerequisites: None

Fundamental concepts of programming using a high-level language. Topics include variables, control structures, functions, arrays, and basic algorithm design. CS 1101 is the entry point for all CS majors and satisfies no prerequisite relationships for 2000-level courses until completed with a grade of C or better. The GPA requirements for CS 1101 specify a minimum grade of C to proceed to CS 1102.

CS 1102 - Object-Oriented Programming

Credit Hours: 3

Prerequisites: CS 1101 with a grade of C or better

Object-oriented design principles, encapsulation, inheritance, polymorphism, and exception handling. The prerequisite relationships for CS 1102 strictly require completion of CS 1101. Students must meet the GPA requirements in CS 1101 (minimum grade of C) to enroll. CS 1102 is a prerequisite for CS 2201, CS 2301, and CS 2401.

CS 2201 - Data Structures and Algorithms

Credit Hours: 3

Prerequisites: CS 1102 with a grade of C or better; MATH 1302 (may be taken concurrently)

Linear and nonlinear data structures including lists, stacks, queues, trees, and graphs. Algorithm analysis using Big-O notation. Sorting and searching algorithms. The prerequisite relationships for CS 2201 require both programming competency from CS 1102 and mathematical maturity from MATH 1302. GPA requirements: minimum grade of C in CS 2201 to enroll in CS 3301 or CS 3302.

CS 2301 - Computer Organization and Architecture

Credit Hours: 3

Prerequisites: CS 1102 with a grade of C or better

Digital logic, assembly language, memory hierarchy, instruction set architecture, and basic processor design. Prerequisite relationships: CS 1102 is required. This course has no direct GPA requirements beyond the general major GPA requirements, but a grade of C or better is recommended before taking CS 3401.

CS 2401 - Discrete Structures for Computer Science

Credit Hours: 3

Prerequisites: CS 1102 with a grade of C or better; MATH 1302 with a grade of C or better

Logic, proof techniques, set theory, combinatorics, graph theory, and discrete probability. Prerequisite relationships require both CS 1102 and MATH 1302. GPA requirements: minimum grade of C required for CS 3301 and CS 3501.

CS 3301 - Analysis of Algorithms

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better; CS 2401 with a grade of C or better; MATH 2301 with a grade of C or better

Advanced algorithm design techniques: divide-and-conquer, dynamic programming, greedy algorithms, amortized analysis, and NP-completeness. The prerequisite relationships for CS 3301 are among the most rigorous in the program, requiring mastery of data structures, discrete mathematics, and calculus. GPA requirements: minimum grade of C required.

CS 3302 - Theory of Computation

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better; CS 2401 with a grade of C or better

Formal languages, automata, computability, and complexity theory. Prerequisite relationships: CS 2201 and CS 2401. GPA requirements: minimum grade of C.

CS 3401 - Operating Systems

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better; CS 2301 with a grade of C or better

Process management, memory management, file systems, concurrency, scheduling, and virtualization. Prerequisite relationships require both CS 2201 and CS 2301. Students must meet the GPA requirements (major GPA ≥ 2.5) to enroll. Graduation rules require completion of CS 3401 or an approved substitute.

CS 3501 - Database Systems

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better; CS 2401 with a grade of C or better

Relational data model, SQL, normalization, transaction processing, concurrency control, and database design. Prerequisite relationships: CS 2201 and CS 2401. GPA requirements: minimum grade of C. This course is required under the graduation rules for all BSCS students.

CS 3601 - Software Engineering

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better

Software development lifecycle, requirements engineering, design patterns, testing, and project management. Prerequisite relationships: CS 2201. GPA requirements: minimum grade of C. CS 3601 is a prerequisite for CS 4901.

CS 3701 - Computer Networks

Credit Hours: 3

Prerequisites: CS 2201 with a grade of C or better; CS 2301 with a grade of C or better

OSI and TCP/IP models, routing, switching, transport protocols, and network security fundamentals. Prerequisite relationships: CS 2201 and CS 2301. GPA requirements: minimum grade of C.

CS 4901 - Senior Capstone I

Credit Hours: 3

Prerequisites: CS 3601 with a grade of C or better; senior standing; major GPA ≥ 2.5

Team-based software development project. Requirements analysis, system design, and project planning. The prerequisite relationships for CS 4901 include CS 3601 and senior standing. GPA requirements are strictly enforced: students must have a major GPA of at least 2.5 to enroll. Graduation rules require both CS 4901 and CS 4902.

CS 4902 - Senior Capstone II

Credit Hours: 3

Prerequisites: CS 4901 with a grade of C or better

Continuation of capstone project. Implementation, testing, deployment, and documentation. Prerequisite relationships: CS 4901. GPA requirements: minimum grade of C. Graduation rules require completion of CS 4902 with a passing grade.

Mathematics Foundation Courses (18 Credit Hours)

Mathematics courses support the theoretical underpinnings of computer science. These courses are subject to the major GPA requirements when used to satisfy CS prerequisites.

MATH 1301 - Calculus I

Credit Hours: 4

Prerequisites: Precalculus or placement exam

Limits, derivatives, integrals, and applications. No CS prerequisite relationships, but required for MATH 1302.

MATH 1302 - Calculus II

Credit Hours: 4

Prerequisites: MATH 1301 with a grade of C or better

Techniques of integration, sequences and series, parametric equations. Prerequisite relationships: MATH 1301. GPA requirements: minimum grade of C required for CS 2201 and CS 2401.

MATH 2301 - Linear Algebra

Credit Hours: 3

Prerequisites: MATH 1302 with a grade of C or better

Vector spaces, matrices, linear transformations, eigenvalues, and eigenvectors. Prerequisite relationships: MATH 1302. GPA requirements: minimum grade of C required for CS 3301.

MATH 2302 - Discrete Mathematics

Credit Hours: 3

Prerequisites: MATH 1302 with a grade of C or better

Advanced combinatorics, recurrence relations, generating functions, and graph algorithms. Prerequisite relationships: MATH 1302. This course is not required for CS majors but may substitute for an elective with advisor approval.

MATH 2401 - Probability and Statistics

Credit Hours: 3

Prerequisites: MATH 1302 with a grade of C or better

Probability theory, random variables, distributions, hypothesis testing, and regression. Prerequisite relationships: MATH 1302. Recommended for students pursuing data science specializations.

General Education Requirements (42 Credit Hours)

General education courses ensure breadth of knowledge. These courses are subject to the cumulative GPA requirements but not the major GPA requirements.

Course Code	Title	Credits	Prerequisites
ENG 1101	College Composition I	3	Placement exam
ENG 1102	College Composition II	3	ENG 1101 with C or better
COMM 1301	Public Speaking	3	ENG 1101 (concurrent allowed)
PHYS 1301	General Physics I	4	MATH 1301 (concurrent allowed)
PHYS 1302	General Physics II	4	PHYS 1301 with C or better

Prerequisite relationships: ENG 1101 is required before ENG 1102. GPA requirements: minimum grade of C in ENG 1101 to enroll in ENG 1102.

Social Sciences (6 credits): Students select two courses from approved social science disciplines. No prerequisite relationships exist for most social science electives. GPA requirements follow the general 2.0 cumulative standard.

Humanities (6 credits): Students select two courses from approved humanities disciplines. No prerequisite relationships exist for most humanities electives. GPA requirements follow the general 2.0 cumulative standard.

Fine Arts (3 credits): One course from approved fine arts offerings. No prerequisite relationships. GPA requirements: general standard.

Additional General Education (10 credits): Remaining general education credits are selected from approved electives. These courses must satisfy the cumulative GPA requirements but do not affect the major GPA requirements.

Prerequisite Relationships

Prerequisite Relationship Matrix

The following table summarizes all prerequisite relationships in the BSCS program. Students must satisfy all prerequisite relationships before registering for a course. Violations of prerequisite relationships may result in administrative withdrawal.

Course	Prerequisites	Min Grade
CS 1102	CS 1101	C
CS 2201	CS 1102, MATH 1302	C
CS 2301	CS 1102	C
CS 2401	CS 1102, MATH 1302	C
CS 3301	CS 2201, CS 2401, MATH 2301	C
CS 3302	CS 2201, CS 2401	C
CS 3401	CS 2201, CS 2301	C
CS 3501	CS 2201, CS 2401	C
CS 3601	CS 2201	C
CS 3701	CS 2201, CS 2301	C
CS 4901	CS 3601, senior standing, major GPA ≥ 2.5	C
CS 4902	CS 4901	C
MATH 1302	MATH 1301	C
MATH 2301	MATH 1302	C
MATH 2302	MATH 1302	C
MATH 2401	MATH 1302	C
PHYS 1302	PHYS 1301	C
ENG 1102	ENG 1101	C

Prerequisite Relationship Enforcement

The registrar's office enforces all prerequisite relationships automatically during registration. Students who believe they have satisfied prerequisite relationships through transfer credit or substitution must submit a Prerequisite Waiver Request to the Computer Science Department. The graduation rules require that all prerequisite relationships be satisfied, even if a waiver is granted for registration purposes.

Chain Prerequisite Relationships

Many courses in the BSCS program form long chains of prerequisite relationships:

- Programming Chain: CS 1101 -> CS 1102 -> CS 2201 -> CS 3301/CS 3302/CS 3401/CS 3501/CS 3601/CS 3701 -> CS 4901 -> CS 4902
- Mathematics Chain: MATH 1301 -> MATH 1302 -> MATH 2301 -> CS 3301
- Systems Chain: CS 1102 -> CS 2301 -> CS 3401/CS 3701

Students should plan their schedules carefully to respect these prerequisite relationships and ensure timely graduation.

GPA Requirements

Cumulative GPA Requirements

The university maintains minimum cumulative GPA requirements for all degree-seeking students. The cumulative GPA requirements for the BSCS program are:

Academic Status	Minimum Cumulative GPA
Good Standing	2.0 or higher
Academic Probation	Below 2.0
Academic Suspension	Below 1.5 for two consecutive semesters

The graduation rules require a cumulative GPA of at least 2.0. Students who fail to meet the cumulative GPA requirements will be placed on academic probation and must restore their GPA to good standing within two semesters.

Major GPA Requirements

The major GPA requirements apply specifically to all courses with CS, MATH, and PHYS prefixes used to satisfy degree requirements. The major GPA requirements are:

Requirement	Minimum Major GPA
Core CS Courses	2.5
Mathematics Foundation	2.5
Technical Electives	2.0
Overall Major	2.5

The graduation rules stipulate that students must achieve a major GPA of at least 2.5 to qualify for graduation. The major GPA requirements are calculated using all attempted major courses, including repeated attempts. The major GPA requirements are separate from and in addition to the cumulative GPA requirements.

GPA Requirements for Specific Courses

Certain courses have individual GPA requirements that function as prerequisite relationships:

- CS 1101: Minimum grade of C required for CS 1102
- CS 1102: Minimum grade of C required for all 2000-level CS courses
- CS 2201: Minimum grade of C required for CS 3301, CS 3302, CS 3401, CS 3501, CS 3601, and CS 3701
- CS 2401: Minimum grade of C required for CS 3301, CS 3302, and CS 3501
- CS 3601: Minimum grade of C required for CS 4901
- CS 4901: Minimum grade of C required for CS 4902; major GPA ≥ 2.5 required for enrollment

These course-level GPA requirements are embedded in the prerequisite relationships and are strictly enforced.

GPA Requirements and Graduation

The graduation rules require satisfaction of all GPA requirements before degree conferral. Students who have completed all credit hours but have not met the GPA requirements must take additional coursework to raise their GPA. The graduation rules do not permit exceptions to the GPA requirements.

Registration and Enrollment Procedures

Registration Timeline

Registration occurs in three phases: priority registration, open registration, and add/drop. Students must resolve all holds and verify that all prerequisite relationships are satisfied before registering.

Course Overrides

Students seeking to enroll in a course without satisfying the prerequisite relationships must obtain a departmental override. Override requests are reviewed by the undergraduate advisor. The graduation rules still require that all prerequisite relationships be satisfied before graduation, even if an override is granted for registration.

Enrollment Limits

Computer Science courses frequently reach enrollment capacity. Students are advised to register early and to have backup schedules prepared. The graduation rules do not extend degree deadlines due to enrollment difficulties.

Academic Standing and Probation

Good Standing

Students are in good standing when they meet all cumulative GPA requirements and major GPA requirements. Good standing is required for priority registration, scholarship eligibility, and participation in departmental activities.

Academic Probation

Students are placed on academic probation when their cumulative GPA falls below 2.0 or their major GPA falls below 2.5. Probationary students must:

- Meet with their academic advisor within the first two weeks of the semester.
- Develop an academic improvement plan.
- Restrict their course load to 12 credit hours or fewer.
- Achieve a semester GPA of at least 2.5 to avoid suspension.

The graduation rules prohibit students on academic probation from enrolling in capstone courses (CS 4901 and CS 4902).

Academic Suspension

Students are subject to academic suspension if their cumulative GPA remains below 2.0 for two consecutive semesters, or their semester GPA falls below 1.5 while on probation.

Suspended students may apply for readmission after one semester. Readmitted students must satisfy all GPA requirements and graduation rules in effect at the time of readmission.

Electives and Specializations

Technical Electives (12 Credit Hours)

Students must complete 12 credit hours of approved technical electives. Technical electives must be 3000-level or above and carry CS, MATH, or approved engineering prefixes. All technical electives are subject to the prerequisite relationships of the specific course and the major GPA requirements.

Course Code	Title	Credits	Prerequisites	Min Grade
CS 4101	Machine Learning	3	CS 3301, MATH 2401	C
CS 4102	Computer Vision	3	CS 3301, MATH 2301	C
CS 4201	Cryptography	3	CS 3302, MATH 2301	C
CS 4202	Network Security	3	CS 3701	C
CS 4301	Compiler Design	3	CS 3301, CS 3302	C
CS 4302	Distributed Systems	3	CS 3401, CS 3701	C
CS 4401	Human-Computer Interaction	3	CS 3601	C
CS 4501	Data Mining	3	CS 3501, MATH 2401	C
CS 4601	Mobile App Development	3	CS 3601	C
CS 4701	Cloud Computing	3	CS 3401	C
MATH 3101	Numerical Analysis	3	MATH 2301, CS 1102	C
MATH 3201	Graph Theory	3	MATH 2302 or CS 2401	C

Prerequisite relationships for all technical electives must be satisfied before enrollment. GPA requirements: minimum grade of C in each technical elective.

Specialization Tracks

Students may optionally concentrate their technical electives in one of the following tracks. Specializations are not required for graduation, but they help structure elective choices.

Artificial Intelligence Track

Required technical electives: CS 4101 - Machine Learning, CS 4102 - Computer Vision, CS 4501 - Data Mining, and one additional CS elective at the 4000 level. Prerequisite relationships for this track emphasize CS 3301, MATH 2301, and MATH 2401. GPA requirements: same as general technical electives.

Cybersecurity Track

Required technical electives: CS 4201 - Cryptography, CS 4202 - Network Security, CS 4302 - Distributed Systems, and one additional CS elective at the 4000 level. Prerequisite relationships for this track emphasize CS 3302 and CS 3701. GPA requirements: same as general technical electives.

Software Engineering Track

Required technical electives: CS 4301 - Compiler Design, CS 4401 - Human-Computer Interaction, CS 4601 - Mobile Application Development, and one additional CS elective at the 4000 level. Prerequisite relationships for this track emphasize CS 3601. GPA requirements: same as general technical electives.

Transfer Credit Policies

Evaluation of Transfer Credits

Transfer credits are evaluated on a course-by-course basis. The graduation rules allow a maximum of 60 transfer credit hours toward the BSCS degree. Transfer courses used to satisfy prerequisite relationships must be equivalent in content to the corresponding courses at this institution.

Transfer GPA Requirements

Transfer credits do not count toward the cumulative GPA requirements or major GPA requirements at this institution. However, students must have achieved a minimum grade of C in any transfer course used to satisfy prerequisite relationships. The graduation rules require that all transfer courses used to satisfy degree requirements be approved by the department.

Residency Requirements

The graduation rules require that students complete at least 30 credit hours at this institution, including CS 4901 and CS 4902. No exceptions to the residency requirements are permitted.

Graduation Application Process

Filing the Graduation Application

Students must file a graduation application through the student portal by the following deadlines:

Graduation Term	Application Deadline
Spring	October 1
Summer	March 1
Fall	July 1

Late graduation applications are subject to a processing fee and may delay degree conferral. The graduation rules require timely filing of the graduation application.

Graduation Audit

Upon receipt of the graduation application, the registrar conducts a graduation audit to verify that all graduation rules have been satisfied. The audit checks:

- Total credit hours (minimum 120)
- Cumulative GPA requirements (minimum 2.0)
- Major GPA requirements (minimum 2.5)
- Completion of all core courses
- Satisfaction of all prerequisite relationships
- Residency requirements (minimum 30 institutional credits)
- Completion of CS 4901 and CS 4902

Students with deficiencies are notified and must resolve them before the graduation audit deadline.

Commencement Participation

Participation in the commencement ceremony requires approval of the graduation application and a preliminary graduation audit confirming that all graduation rules are expected to be satisfied by the end of the term. Students who fail to meet the GPA requirements or other graduation rules after participating in commencement will not have their degree conferred until all deficiencies are resolved.

Post-Graduation GPA Requirements

Once the degree is conferred, the transcript is finalized. The graduation rules and GPA requirements are permanently sealed. Students may not retroactively improve their GPA or repeat courses to alter their graduation status.

Summary of Key Academic Requirements

Graduation Rules Checklist

To ensure compliance with all graduation rules, students should verify the following before filing their graduation application:

- ☐ 120 total credit hours completed
- ☐ All general education requirements satisfied
- ☐ All CS core courses completed with grade of C or better
- ☐ All mathematics foundation courses completed with grade of C or better
- ☐ 12 credit hours of technical electives completed
- ☐ All prerequisite relationships satisfied
- ☐ Cumulative GPA ≥ 2.0
- ☐ Major GPA ≥ 2.5
- ☐ 30 credit hours of residency completed
- ☐ CS 4901 and CS 4902 completed
- ☐ Graduation application filed by deadline

GPA Requirements Summary

GPA Type	Minimum Requirement	Courses Included
Cumulative GPA	2.0	All attempted coursework
Major GPA	2.5	All CS, MATH, and PHYS prefix courses
Course-Level GPA	C (2.0)	Individual prerequisite courses

The graduation rules require satisfaction of all three categories of GPA requirements. No waivers are granted for GPA requirements.

Academic Integrity and Conduct

Honor Code

All students are bound by the university honor code. Violations of academic integrity, including plagiarism, cheating, and unauthorized collaboration, are reported to the Office of Student Conduct. Sanctions may include grade penalties, course failure, suspension, or expulsion. The graduation rules specify that students dismissed for academic integrity violations forfeit all progress toward the degree.

Grade Appeals

Students may appeal final grades through the established grade appeal process. Grade appeals must be initiated within

30 days of grade posting. Successful grade appeals may affect cumulative GPA requirements and major GPA requirements, but cannot override the graduation rules regarding credit hours or prerequisite relationships.

Advising and Student Support

Academic Advising

All BSCS students are assigned a faculty advisor upon declaration of the major. Students must meet with their advisor at least once per semester to discuss course selection, prerequisite relationships, and progress toward graduation. Advisors help students understand the graduation rules and GPA requirements.

Tutoring and Academic Support

The university provides free tutoring for CS 1101, CS 1102, CS 2201, MATH 1301, and MATH 1302. Students struggling to meet the GPA requirements are strongly encouraged to utilize these resources.

Career Services

The Career Services Center assists students with internship placement, resume preparation, and job search strategies. Internships do not satisfy graduation rules or degree requirements unless completed as part of an approved cooperative education program.

Appendices

Appendix A: Course Code Prefix Key

Prefix	Department
CS	Computer Science
MATH	Mathematics
PHYS	Physics
ENG	English
COMM	Communication

Appendix B: Glossary of Terms

Cumulative GPA:

The grade point average calculated across all attempted coursework at the institution.

Major GPA:

The grade point average calculated across all courses in the major discipline.

Prerequisite Relationships:

The academic requirements that must be satisfied before enrolling in a specific course.

Graduation Rules:

The comprehensive set of requirements that must be satisfied to earn a degree.

GPA Requirements:

The minimum grade point averages that students must maintain to remain in good standing and qualify for graduation.

Residency Requirements:

The minimum number of credit hours that must be completed at the degree-granting institution.

Technical Electives:

Advanced courses selected from an approved list to deepen knowledge in specialized areas.

Appendix C: Contact Information

Office	Location	Phone
Computer Science Department	Science Building, Room 301	(555) 123-4567
Registrar's Office	Administration Building, Room 102	(555) 123-4568
Academic Advising Center	Student Union, Room 205	(555) 123-4569
Financial Aid Office	Administration Building, Room 110	(555) 123-4570

This handbook is effective for the 2026-2027 academic year. The university reserves the right to modify graduation rules, GPA requirements, and prerequisite relationships. Students are responsible for consulting the most current version of this handbook and working with their academic advisor to ensure compliance with all degree requirements.